N5

NATIONAL

QUALIFICATIONS

Mark

N5/K

Mathematics

Combined Paper

45 minutes

Prelim Practice K

Fill in these boxes and read what is printed below

Forename(s)

Surname

Teacher

Total Marks - 30

Attempt ALL questions.

You must NOT use a calculator.

To earn full marks you must show your working in your answers.

State the units for your answer where appropriate.

Write your answers clearly in the spaces provided in this booklet. Additional space for answers is provided at the end of this booklet. If you use this space you must clearly identify the question number you are attempting.

Use **blue** or **black** ink.

Before leaving the examination room you must give this book to the Invigilator; if you do not, you may lose all the marks for this paper.

FORMULAE LIST

The roots of $ax^2 + bx + c = 0$ are $x = \dfrac{-b \pm \sqrt{(b^2 - 4ac)}}{2a}$

Sine rule: $\dfrac{a}{\sin A} = \dfrac{b}{\sin B} = \dfrac{c}{\sin C}$

Cosine rule: $a^2 = b^2 + c^2 - 2bc \cos A$ or $\cos A = \dfrac{b^2 + c^2 - a^2}{2bc}$

Area of a triangle: $A = \frac{1}{2} ab \sin C$

Volume of a sphere: $V = \dfrac{4}{3}\pi r^3$

Volume of a cone: $V = \dfrac{1}{3}\pi r^2 h$

Volume of a pyramid: $V = \dfrac{1}{3} Ah$

Standard deviation: $s = \sqrt{\dfrac{\sum (x - \bar{x})^2}{n - 1}}$

or $s = \sqrt{\dfrac{\sum x^2 - \dfrac{(\sum x)^2}{n}}{n - 1}}$, where n is the sample size.

MARKS

Section A – NON-CALCULATOR
Total marks — 14
Attempt ALL questions

1. Simplify:

$$\frac{2x-5}{3}+\frac{x-2}{4}$$

2

2. Given $\boldsymbol{u}=\begin{pmatrix}-1\\-5\\2\end{pmatrix}$ and $\boldsymbol{v}=\begin{pmatrix}7\\-6\\6\end{pmatrix}$, find the magnitude of $|\boldsymbol{v}-\boldsymbol{u}|$. **3**

3. a) Express $x^2-14x+45$ in the form $(x+a)^2+b$. **2**

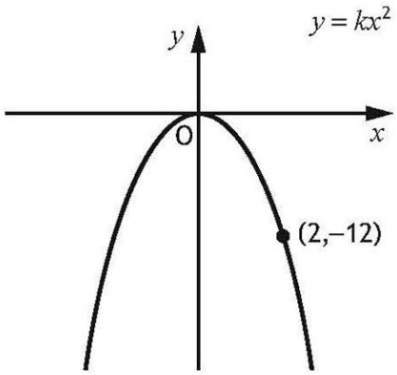
b) State the coordinates of the turning point on the graph of $y=x^2-14x+45$. **1**

4. Express as a single fraction in its simplest form, where $x \neq 5$:

$$\frac{4}{(x-5)^2} \div \frac{7}{(x-5)^3}$$

2

5. The diagram below shows part of the graph of $y = kx^2$.



Find the value of k. **2**

6. Simplify: $5 \tan x \cos x$ **2**

[END OF SECTION A]

Section B – CALCULATOR ALLOWED
Total marks — 16
Attempt ALL questions

7. Solve the equation $2x^2 + 6x - 3 = 0$.
Give your answer correct to **three significant figures**. **4**

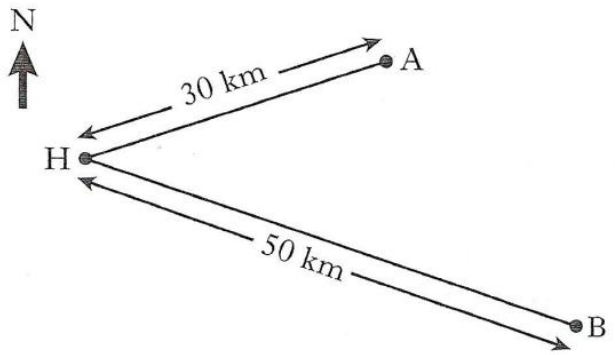
8. Solve the equation $3 + 7\tan x° = 6$ where $0 < x < 360$. **3**

9. Determine the nature of the roots of the function $f(x) = 3x^2 - 6x + 3$. **2**

10. Two yachts leave from harbour H.

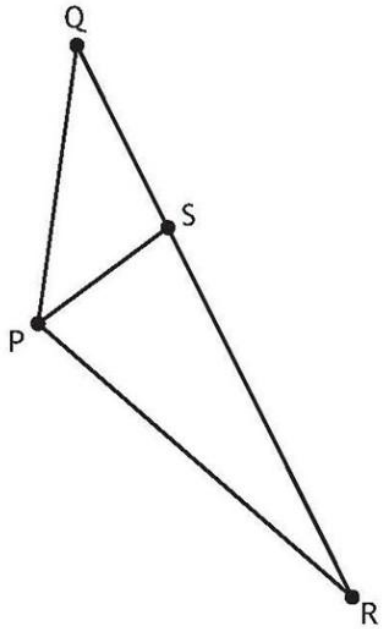
Yacht A sails on a bearing of 072° for 30 kilometres.

Yacht B sails on a bearing of 140° for 50 kilometres.



Calculate how far apart the two yachts are. **4**

11. In the diagram, $\overrightarrow{RP} = \boldsymbol{u}$ and $\overrightarrow{RQ} = \boldsymbol{v}$.



a) Express $\overrightarrow{PQ}$ in terms of $\boldsymbol{u}$ and $\boldsymbol{v}$. **1**

$\overrightarrow{SQ} = \frac{1}{2}\overrightarrow{RS}$.

b) Express $\overrightarrow{PS}$ in terms of $\boldsymbol{u}$ and $\boldsymbol{v}$. **2**

[END OF SECTION B]

ANSWERS

1. $\frac{11x-26}{12}$

2. 9

3. a) $(x-7)^2-4$ b) $(7,-4)$

4. $\frac{4(x-5)}{7}$ or $\frac{4x-20}{7}$

5. $k=-3$

6. $5\sin x$

7. $x=0.436, x=-3.44$

8. $23.2°$ and $203.2°$

9. $b^2-4ac=0$ so real equal roots

10. 47.7 km

11. a) $-\boldsymbol{u}+\boldsymbol{v}$ b) $-\boldsymbol{u}+\frac{2}{3}\boldsymbol{v}$